S3.2 Functional groups: Classification of organic compounds

Infrared (IR) spectroscopy (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to interpret the functional group region of an IR spectrum, using a table of characteristic frequencies (wavenumber cm^{-1}).

Have you ever sat inside a car on a hot sunny day and experienced the high temperature inside? Temperatures inside a car can reach up to 50 °C, which can prove fatal. But why does it get that hot? Light from the Sun consists of mainly short wavelength radiation (visible and ultraviolet radiation). The light passes through the windows of the car and is absorbed by the interior. The interior of the car re-radiates some of this light as longer wavelength infrared radiation, which cannot pass out through the windows and has the effect of heating up the inside of the car. The same effect occurs in a greenhouse as well as on the planet Earth and is known as the greenhouse effect (**Figure 1**).

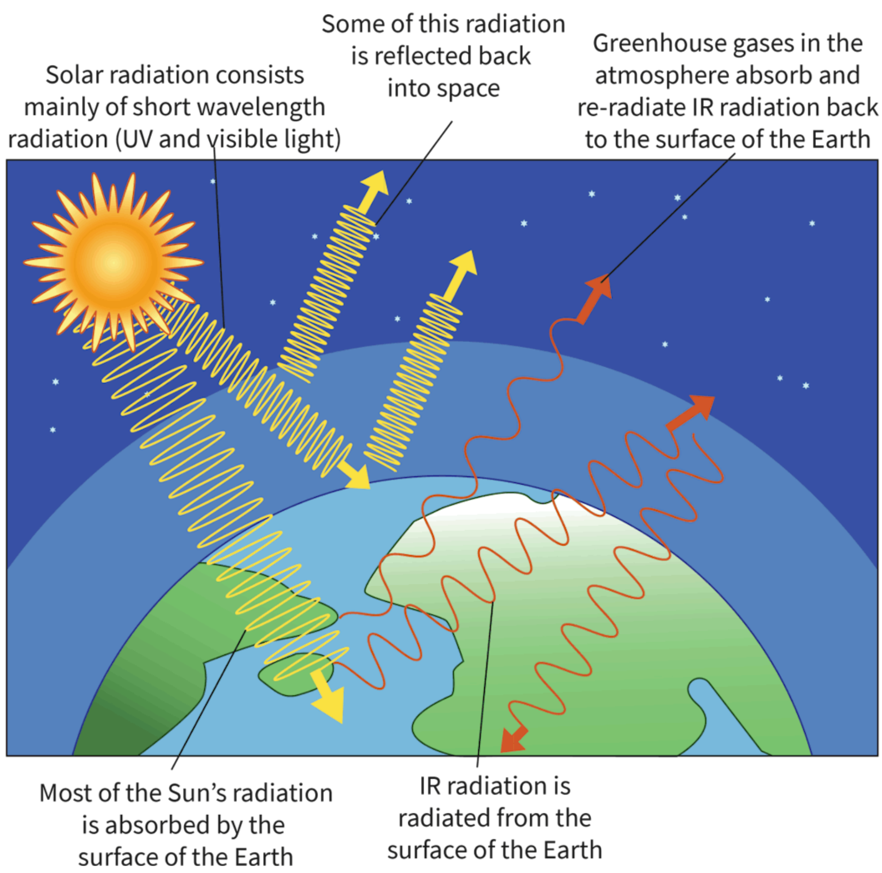

Figure 1. Greenhouse gases are responsible for the greenhouse effect on Earth.

More information for figure 1

The diagram illustrates the greenhouse effect on Earth. It shows the Sun emitting solar radiation, which consists mainly of short wavelength radiation like UV and visible light. This radiation passes through the Earth's atmosphere and is mostly absorbed by the Earth's surface. Some of the solar radiation is reflected back into space. The Earth's surface then radiates infrared (IR) radiation. Greenhouse gases present in the atmosphere absorb and re-radiate this IR radiation back towards the Earth's surface, contributing to warming. Arrows are used to indicate the direction of radiation flow, highlighting how solar radiation enters and infrared radiation is trapped in the Earth's atmosphere.

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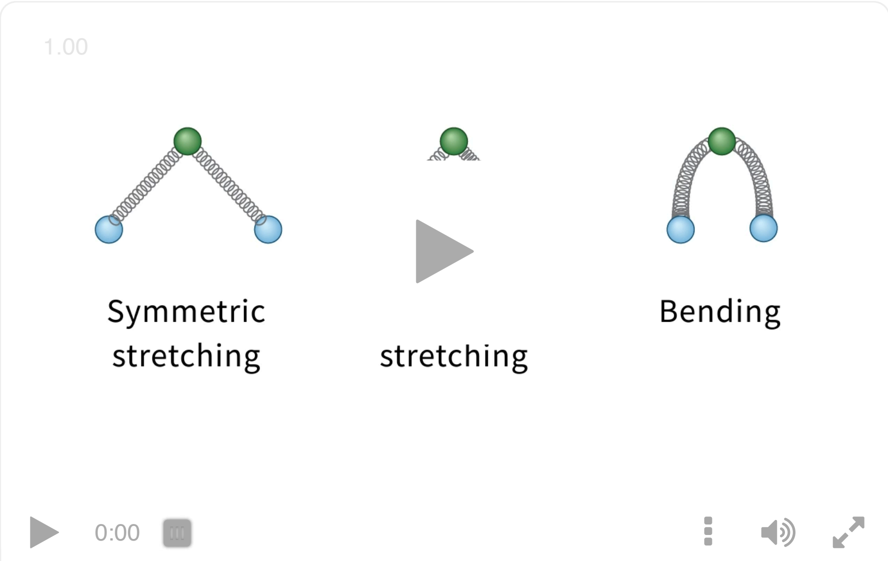
Infrared spectroscopy is a technique that uses infrared radiation (see section S1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/)) to determine the type of bonds in a compound. Chemical bonds can be thought of as behaving like springs between atoms, due to the movement of the electrons. Rather than being rigid, they vibrate at specific frequencies. If they are exposed to electromagnetic radiation of the same frequency at which the bond vibrates, the bond will absorb the radiation and vibrate. This forms the basis of the technique of infrared spectroscopy (IR spectroscopy). Different bonds absorb different amounts of infrared radiation. They can also vibrate in different ways. These fall under two basic categories, which are stretches and bends. The ways in which a bond can vibrate, known as modes of vibration or vibrational modes, are summarised in **Interactive 1**.



Interactive 1. The Different Modes of Vibration of a Molecule: Stretching and Bending.

More information for interactive 1

The animation video illustrates the different modes of molecular vibration. There are 3 different molecules in the video labeled symmetric stretching, asymmetric stretching, and bending. Each molecule features a central green sphere bonded to two blue spheres that are positioned at the bottom left and bottom right of the green sphere. These spheres represent atoms in a molecule.
In this case, the bonds are represented by springs, visually demonstrating the vibrational movements.
In the symmetric stretching, both blue spheres move toward and away from

the central green sphere in sync. The bond lengths between the green and blue spheres increase or decrease together in synchronization.
In asymmetric stretching, when one blue sphere moves toward the green sphere, the other blue sphere moves away from the green sphere. In this stretching, the bond length between one blue sphere and the green sphere increases, while the bond length between the other blue sphere and the green sphere decreases at the same time.
In the bending, both blue spheres move toward and away from each other like a pair of scissors. The bond length remains unchanged, but the bond angle changes.
This video helps users understand the different modes of molecular vibration and how these vibrations affect bond length and bond angles.

By measuring the amount of infrared radiation absorbed by the bonds in an organic compound, an infrared spectrum is produced. Using this spectrum, together with a list of regions and corresponding structures such as those found in section 20 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/infrared-data-id-45122/) of the DP chemistry data booklet, we can determine which bonds are present in the molecule and therefore which functional groups are present. **Table 1** shows the type of bond together with the wavenumber at which it vibrates. The wavenumber is the reciprocal of wavelength $\left(\frac{1}{\lambda}\right)$ and has the unit cm^{-1}.

Table 1. Types of bond and their vibrations.

Bond	Types of organic molecules	Wavenumber / cm^{-1}	Intensit
C—I	Iodoalkanes	490—620	Strong
C—Br	Bromoalkanes	500—600	Strong
C—Cl	Chloroalkanes	600—800	Strong
C—F	Fluoroalkanes	1000—1400	Strong
C—O	Alcohols, esters, ethers	1050—1410	Strong
C=C	Alkenes	1620—1680	Medium-w multiple ba
C=O	Aldehydes, ketones, carboxylic acids and esters	1700—1750	Strong
C≡C	Alkynes	2100—2260	Variable
O—H	Carboxylic acids (with hydrogen bonding)	2500—3000	Strong, very broa

Bond	Types of organic molecules	Wavenumber / cm^{-1}	Intensit
C—H	Alkanes, alkenes, arenes	2850—3090	Strong
O—H	Alcohols and phenols (with hydrogen bonding)	3200—3600	Strong, br
N—H	Primary amines	3300—3500	Mediur two banc

Study skills

The final column in **Table 1** is 'intensity', which is determined by the polarity of the bond. The intensity depends on the dipole moment of the bond. Strongly polar bonds produce strong bands, bonds with medium polarity produce medium bands and weakly polar bonds produce weak bands.

An infrared spectrum shows the percentage transmittance on the *y*-axis and wavenumber on the *x*-axis. The percentage transmittance is the percentage of the IR radiation that is transmitted. Values of less than 100% mean that IR radiation is being absorbed and a downwards pointing peak is produced. These peaks are known as either absorptions or stretches.

The two main regions of an infrared spectrum are the fingerprint region and the functional group region (**Figure 2**). The fingerprint region is located below a wavenumber of 1500 cm^{-1}. This area is unique to each compound and can be used to identify an unknown compound by comparing this region with the fingerprint region of a known compound. The functional group region (from 1500 cm^{-1} upwards) is used to determine the types of bond present in the molecule. It is this region that we will focus on for the remainder of the section.

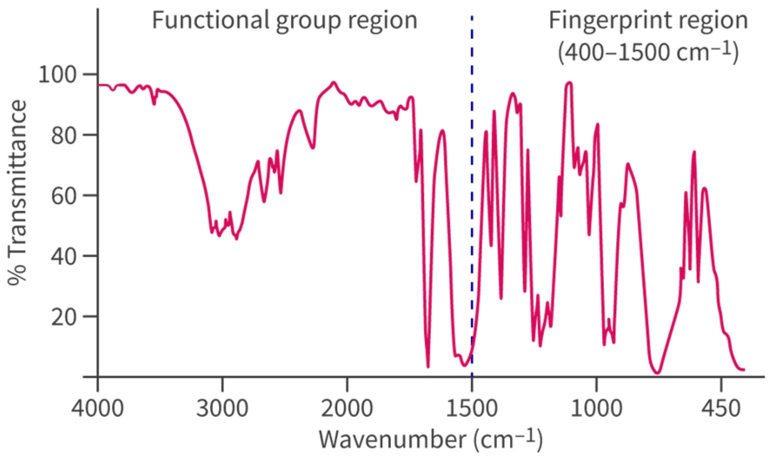


Figure 2. An infrared spectrum can be divided into two main regions: the fingerprint region and the functional group region.

More information for figure 2

The image displays an infrared spectrum graph with 'Wavenumber (cm−1)' on the X-axis ranging from 400 to 4000 cm−1 and '% Transmittance' on the Y-axis ranging from 0 to 100. The graph has two main labeled areas: the 'Functional group region' on the left, and the 'Fingerprint region (400—1500 cm−1)' on the right. These labels are marked above the graph. The graph line itself shows several peaks and valleys, indicating various absorbance features relevant to different chemical bonds. The fingerprint region, specifically, has a complex pattern of peaks that can uniquely identify specific compounds.

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Figure 3 shows the IR spectrum of ethanol, CH_3CH_2OH. The spectrum shows two stretches (or absorptions) in the functional group region, one at approximately 2900 cm^{-1} and the other between 3200 and 3400 cm^{-1}. Referring to **Figure 3**, we can see that the stretch at 2900 cm^{-1} is characteristic of a C–H bond. The strong and broad stretch at between 3200 and 3400 cm^{-1} is characteristic of an O–H bond in alcohols and phenols caused by hydrogen bonding. This is a very distinctive stretch and is an easy way to identify an alcohol.

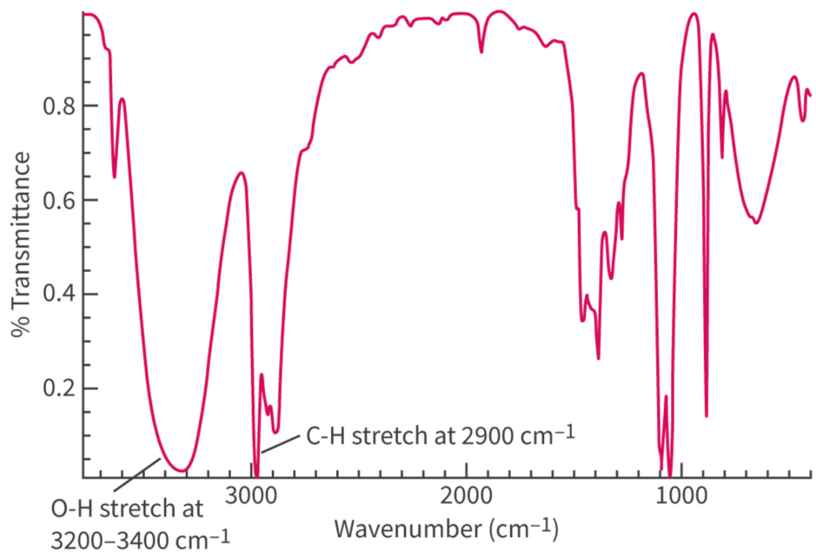

Figure 3. The infrared spectrum of ethanol shows a distinctive stretch between 3200 and 3400 cm^{-1}.

More information for figure 3

The image shows an infrared spectrum graph of ethanol with transmittance on the Y-axis and wavenumber on the X-axis. The X-axis ranges from 400 cm−1 to 3600 cm−1. Two significant stretches are labeled: one at 2900 cm−1 indicating a C-H bond, and a broader, strong stretch between 3200 cm−1 and 3400 cm−1, indicating an O-H bond due to hydrogen bonding. These stretches help identify ethanol's functional groups, distinguishing the alcohol's properties.

[Generated by AI]

Our next example is the infrared spectrum of ethanoic acid, CH_3COOH in **Figure 4**. Again, we see two distinctive stretches on the spectrum: one at 1750 cm^{-1} and the other between 2800 and 3200 cm^{-1}. The stretch at 1750 cm^{-1} is characteristic of a C=O bond found in a variety of different organic compounds such as aldehydes, ketones, carboxylic acids and esters. The strong and very broad stretch at between 2800 and 3200 cm^{-1} is characteristic of an O–H bond. However, this absorption differs in shape from the one we observed in the IR spectrum of ethanol in **Figure 3** as the O–H bond in carboxylic acid exists in a different environment due to the close proximity of the C=O bond. When we observe both these stretches on an IR spectrum, we can deduce that the molecule contains a carboxyl functional group, COOH. Note that there are also C–H bonds in ethanoic acid that absorb IR radiation but the stretch is obscured by the O–H stretch.

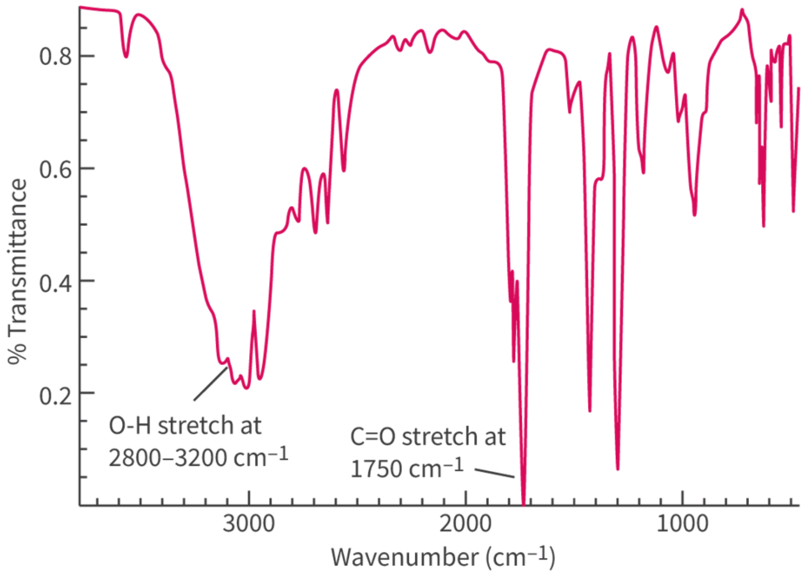

Figure 4. The infrared spectrum of ethanoic acid shows two distinctive stretches: one at 1750 cm^{-1} and one between 2800 and 3200 cm^{-1}.

More information for figure 4

The image is a graph displaying the infrared spectrum of ethanoic acid. The Y-axis represents % Transmittance, with a scale ranging from 0 to 1. The X-axis represents Wavenumber in cm^{-1}, ranging from 4000 to 500. There are two distinctive stretches marked on the graph. The first is a broad O-H stretch between 2800 and 3200 cm^{-1}, characterized by a peak in transmittance. The second is a sharp C=O stretch at 1750 cm^{-1}. Both stretches are key indicators of the carboxylic acid functional group in ethanoic acid, with the O-H bond absorbing differently due to the proximity of the C=O bond. Various other peaks and troughs are also displayed across the spectrum, reflecting additional bonds such as C-H.

[Generated by AI]

The last example we will look at is the IR spectrum of ethanal, CH_3CHO in **Figure 5**. The stretch at 1750 cm^{-1} is characteristic of a C=O bond, which can be found in a variety of different organic compounds. The absence of a stretch can sometimes be used to determine the functional group: in this spectrum the absence of the stretch between 2500 and 3000 cm^{-1} tells us that this compound cannot be a carboxylic acid.

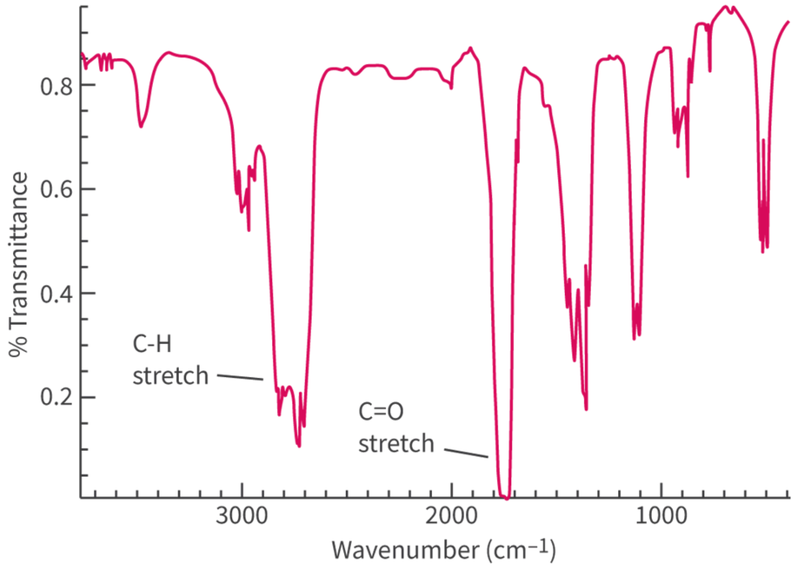

Figure 5. The infrared spectrum of ethanal shows a distinctive stretch at 1750 cm^{-1}.

More information for figure 5

The image is a graph showing the infrared (IR) spectrum of ethanal. The X-axis represents the wavenumber in cm−1, ranging from approximately 3500 to 500 cm−1. The Y-axis displays % transmittance, ranging from 0 to 1, where 1 represents 100% transmittance.

The graph depicts several peaks and valleys indicating different molecular vibrations. Notably, there is a steep drop at 1750 cm−1, marking the C=O stretch, which is a characteristic peak identifying the presence of a carbonyl group within the molecule. Another labeled region is a smaller dip around 2900 cm−1, which is associated with C-H stretching.

These features provide information about the structural elements within the ethanal molecule, which can be used to deduce that the absence of certain bands, such as those between 2500 and 3000 cm−1, indicates the compound cannot be a carboxylic acid. This explanation is vital for understanding the compound's functional group characteristics.

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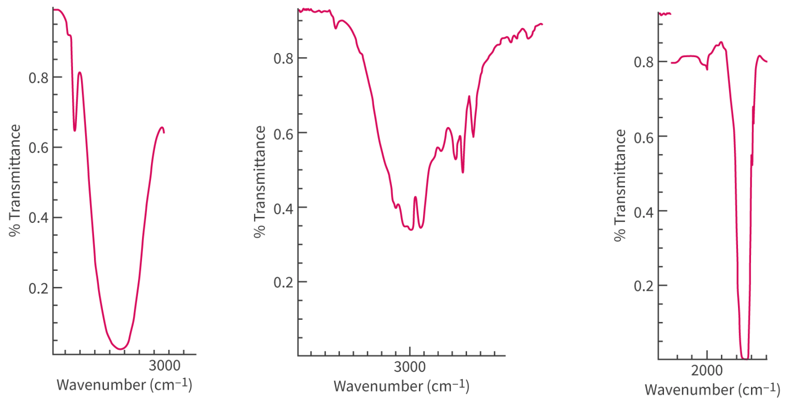
Drag and drop the type of stretch to match the correct bond in **Interactive 2**.



C=O	O–H (carboxylic acids)	O–H (alcohols)

Check

Interactive 2. Match the correct bond to the correct stretch.

More information for interactive 2

This drag and drop interactive displays three infrared spectra, each plotting transmittance percentage on the y-axis and wave number (cm^{-1}) on the x-axis.

Each spectrum contains a notable dip, indicating the absorption bands of specific functional groups:

Spectrum 1: A strong, broad dip between 3300 and 3400 cm-1, reaching 0.03% transmittance
Spectrum 2: A strong, very broad dip between 2900 and 3100 cm-1, reaching 0.35% transmittance
Spectrum 3: A strong dip at 1750 cm-1, reaching 0% transmittance

All values are approximate.

This interactive tool helps users identify infrared absorption regions corresponding to different functional groups.

Read below for the solution:

Spectrum 1: O–H (alcohols)
Spectrum 2: O–H (carboxylic acids)
Spectrum 3: $\text{C} = \text{O}$

IR absorption by greenhouse gases

Greenhouse gases are gases that absorb infrared radiation and contribute to the greenhouse effect, which was discussed at the beginning of the section. But what makes a gas a greenhouse gas? In other words, why do some gases absorb IR radiation and some do not? The answer lies in the dipole moment of the molecule, more specifically the change in the dipole moment of the molecule. Take carbon dioxide (CO_2), for example, which is a potent greenhouse gas. Two modes of vibration of the CO_2 molecule are shown in **Figure 6**.

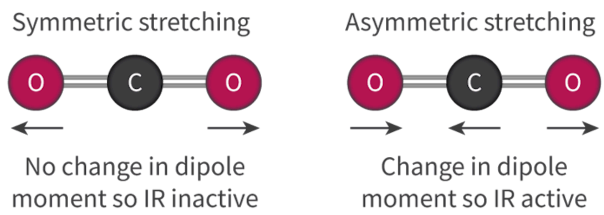


Figure 6. Two possible modes of vibration for CO_2: symmetric stretching, which is IR inactive and asymmetric stretching, which is IR active.

More information for figure 6

The diagram illustrates two modes of vibration for a carbon dioxide (CO2) molecule. On the left side, it shows symmetric stretching, where the oxygen atoms move in and out at the same rate from the center carbon atom. Underneath, it states that there is 'No change in dipole moment so IR inactive'. On the right side, it depicts asymmetric stretching, where the oxygen atoms move at different rates, causing a change in the molecule's dipole moment. The text under this illustration reads 'Change in dipole moment so IR active'. The diagram emphasizes the different interactions regarding infrared absorption due to the change in dipole moments.

[Generated by AI]

CO_2 is a non-polar molecule because the bond dipoles cancel out. When the CO_2 molecule undergoes symmetrical stretching, there is no change in the dipole moment of the molecule, therefore it does not absorb IR radiation and is known as being IR inactive. However, when the CO_2 molecule undergoes asymmetric stretching there is a change in the dipole moment of the molecule because the bond dipoles no longer cancel out. This change in the dipole moment means that the CO_2 can now absorb IR radiation and is known as being IR active. Note that to absorb IR radiation, a molecule does not have to have a permanent dipole (such as CO_2) but there must be a change in the dipole moment of the molecule.

Making connections

What properties of a greenhouse gas determine its ‘global warming potential’?

In subtopic R1.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/), you learned about the relationship between carbon dioxide levels in the atmosphere and the greenhouse effect.

The next greenhouse gas we will look at is water, H_2O. Three vibrational modes of water are shown in **Figure 7**, which are all IR active. You may recall that water is a polar molecule in its ground state but because each mode results in a change in the dipole moment of the molecule, all three modes are IR active.

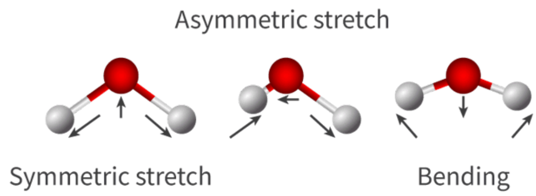


Figure 7. The three vibrational modes of H_2O, which are all IR active.

More information for figure 7

The image is a diagram illustrating the three vibrational modes of the water molecule (H2O). The first mode shown is the asymmetric stretch, where the oxygen atom is in the center with two hydrogen atoms on either side. Arrows indicate that one hydrogen is moving towards and the other away from the oxygen atom. The second mode is the symmetric stretch, where both hydrogen atoms move synchronously towards and away from the oxygen atom, shown by arrows pointing towards and away from the oxygen atom. The final mode is bending, where the two hydrogen atoms move in a way that makes the angle between them smaller and larger, indicated by arrows showing inward and outward movement. The text labels for each mode accompany the illustrations above or below each depiction.

[Generated by AI]

The last greenhouse gas we will look at is methane. Like CO_2, it is a nonpolar molecule however, when it interacts with infrared radiation, there is a change in the dipole moment of the molecule. This change in the dipole moment allows methane to absorb infrared radiation and act as a greenhouse gas. For the methane molecule, there are two vibrational modes that are IR active; these are asymmetric stretching and bending.

Diatomic molecules such as N_2 and O_2, despite making up approximately 99% of the Earth's atmosphere, do not act as greenhouse gases. Oxygen gas and nitrogen gas have only one vibrational mode, which involves a stretching and compression of the bond lengths (**Figure 8**). There is no change in the dipole moment of the molecule so they do not absorb infrared radiation and are IR inactive.

Figure 8. O_2 and N_2 only have one vibrational mode, which does not create a change in the dipole moment of the molecule.

More information for figure 8

The image illustrates two diatomic molecules. On the left, there is an oxygen molecule (O2) consisting of two red spheres, each labeled 'O,' connected by a double bond. On the right, there is a nitrogen molecule (N2) consisting of two blue spheres, each labeled 'N,' connected by a triple bond. These representations demonstrate the molecular structure of these gases, emphasizing their single vibrational modes due to their double and triple bonds, respectively. These vibrational modes do not change their dipole moments, making them IR inactive, which relates to the text describing why these gases do not function as greenhouse gases.

[Generated by AI]

In this activity, you will interpret three infrared spectra to determine the bonds and functional groups present.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills —Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. Determine the bonds responsible for the following stretches.

(a) The broad stretch at 2800—3200 cm^{-1}.
(b) The stretch at 1750 cm^{-1}.
(c) The functional group of the molecule.

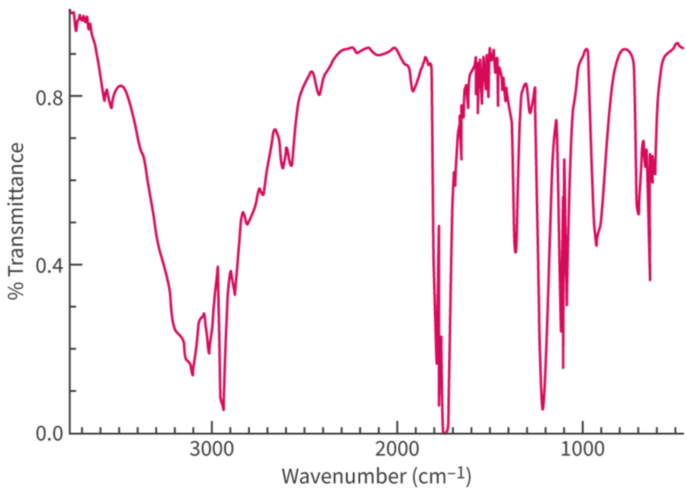


Figure 9. Determine the bonds.

More information for figure 9

The image is an infrared spectrum graph with transmittance on the Y-axis and wavenumber in cm^-1 on the X-axis. The transmittance scale on the Y-axis ranges from 0.0 to 1.0. The X-axis displays wavenumber values from around 3500 cm^-1 to 500 cm^-1.

Several key features are observable in the graph: 1. A broad, shallow stretch is visible between 2800—3200 cm^-1, likely corresponding to O-H or N-H stretching vibrations. 2. A sharp peak is present at approximately 1750 cm^-1, which is characteristic of a C=O carbonyl stretch. 3. Additional sharp peaks are present at lower wavenumbers, indicating other functional group vibrations of the molecule.

This spectral analysis enables identification of functional groups present in the molecule by correlating peaks with known absorption frequencies.

[Generated by AI]

2. Determine the bonds responsible for the following stretches.

(a) The broad stretch at 3300—3400 cm^{-1}.
(b) The stretch at 2950 cm^{-1}.
(c) The functional group of the molecule.

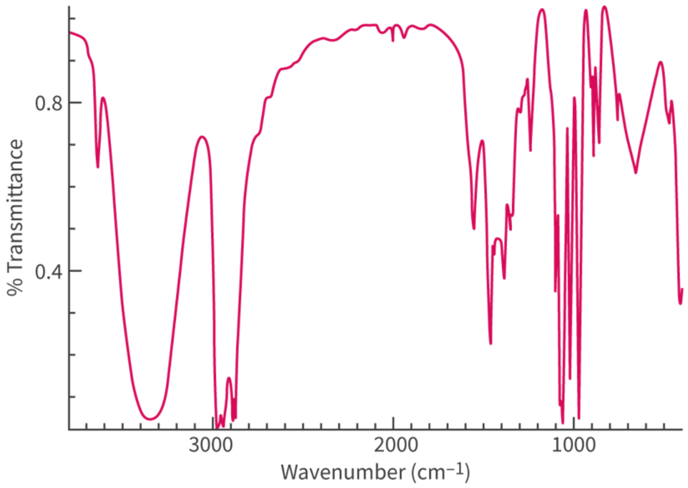


Figure 10. Determine the bonds.

More information for figure 10

The image is a spectral graph representing the percentage of transmittance (% Transmittance) on the Y-axis and the wavenumber in cm−1 (Wavenumber (cm−1)) on the X-axis. The graph shows multiple peaks and valleys of varying shapes and amplitudes across the spectrum. Key features include a broad stretch between 3300 and 3400 cm−1, a narrower peak around 2950 cm−1, and various additional peaks at lower wavenumbers. The broad stretch and peaks likely indicate different bonds and functional groups within a molecule. The data trend shows significant absorbance at specific ranges, pertinent to chemical bond identification.

[Generated by AI]

3. Determine the bonds responsible for the following stretches.

(a) The stretch at 1750 cm^{-1}.
(b) The stretch at 2950 cm^{-1}.
(c) The functional group of the molecule.

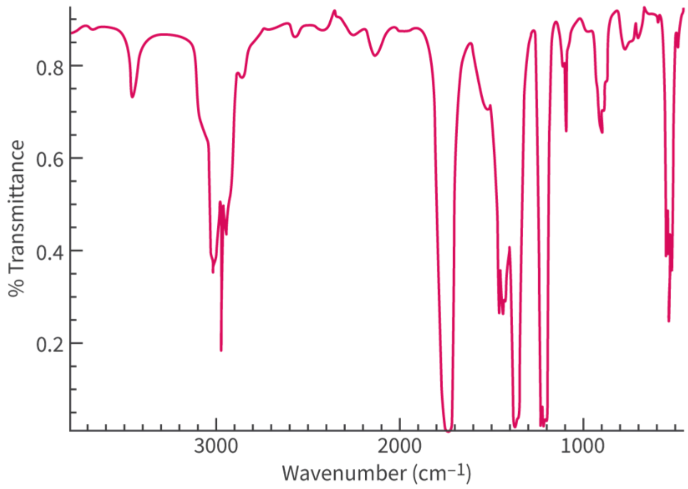


Figure 11. Determine the bonds.

More information for figure 11

The image is a graph of infrared spectroscopy data, which features a plot line showing the % Transmittance on the Y-axis against the Wavenumber (in cm−1) on the X-axis. The X-axis ranges from over 3000 cm−1 to under 500 cm−1, while the Y-axis displays transmittance levels, marked from 0 to 1. The plot line, which appears in magenta, demonstrates several peaks and valleys, indicating various vibrational bonds. Notable points include valleys at approximately 1750 cm−1 and 2950 cm−1, which might suggest specific bond stretches. These features help in identifying functional groups present in the sample being analyzed, as specific bonds absorb characteristic frequencies of IR radiation, creating peaks and troughs in the spectrum.

[Generated by AI]

1. (a) O—H bond in carboxylic acids due to hydrogen bonding.
   (b) C=O in aldehydes, ketones, carboxylic acids and esters.
   (c) The functional group is a carboxyl group, COOH.
2. (a) O—H bond in alcohols due to hydrogen bonding.
   (b) C—H bond.
   (c) The functional group is a hydroxyl group.
3. (a) C=O in aldehydes, ketones, carboxylic acids and esters.
   (b) C—H bond.
   (c) The functional group is a carbonyl group. It cannot be a carboxyl group because of the absence of the stretch at 3300—3400 cm^{-1}.